

Reasoning: Calendar - Complete Study Material

Theory, 15 Solved Examples, and 25 Practice Questions with Answer Key

Part 1: Theory (Important Rules of Calendar)

1. Types of Years

- **Ordinary Year:** Contains 365 days. It has 52 weeks and 1 odd day ($365 \div 7$ remainder = 1).
- **Leap Year:** Contains 366 days (February has 29 days instead of 28). It has 52 weeks and 2 odd days ($366 \div 7$ remainder = 2).
- **Identifying a Leap Year:** Any year divisible by 4 is a leap year. For century years (like 1700, 1900, 2000), they must be divisible by 400.

2. Concept of Odd Days

The remainder left after dividing the total number of days by 7 is called an **Odd Day**.

- Number of odd days in 100 years = 5
- Number of odd days in 200 years = 3
- Number of odd days in 300 years = 1
- Number of odd days in 400 years = 0

3. Day Codes

Odd Days	0	1	2	3	4	5	6
Day	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday

4. Odd Days in Months

January (3), February (Ordinary: 0, Leap: 1), March (3), April (2), May (3), June (2), July (3), August (3), September (2), October (3), November (2), December (3).

Part 2: Solved Examples (15 Examples)

Example 1: What day of the week was 15th August 1947?

Solution: Up to 1947: 1600 years (0) + 300 years (1) + 46 years ($46 + 11 = 57 \circ 1$ odd day). Days in 1947 up to 15 August = $(3+0+3+2+3+2+3) + 15 = 16 + 15 = 31$ days $\circ 31 \div 7$ remainder 3. Total odd days = $1 + 3 = 4$.

Answer: Thursday

Example 2: If 1st January 2004 was Thursday, what day will it be on 1st January 2005?

Solution: The year 2004 is a leap year, so it has 2 odd days. Adding 2 days to Thursday gives Saturday.

Answer: Saturday

Example 3: If the first day of an ordinary year is Tuesday, what will be the last day of that year?

Solution: The first and last day of an ordinary year are always the same. If the first day is Tuesday, the last day is also Tuesday.

Answer: Tuesday

Example 4: If the first day of a leap year is Monday, what will be the last day of that year?

Solution: The last day of a leap year is one day ahead of its first day. Monday + 1 = Tuesday.

Answer: Tuesday

Example 5: The calendar for the year 2012 will be the same for which year?

Solution: A leap year calendar repeats after 28 years. $2012 + 28 = 2040$.

Answer: Year 2040

Example 6: The calendar for the year 2017 will be the same for which year?

Solution: The calendar for the first year after a leap year repeats after 6 years. $2017 + 6 = 2023$.

Answer: Year 2023

Example 7: Which of the following cannot be the last day of a century?

Solution: A century can never end on Tuesday, Thursday, or Saturday.

Answer: Tuesday / Thursday / Saturday

Example 8: What day of the week was 26th January 1950?

Solution: Calculating the total odd days up to 26 January 1950 gives Thursday.

Answer: Thursday

Example 9: If 5th of a month is Tuesday, what day will be 3 days before the 25th of that month?

Solution: 3 days before 25th = 22nd. Difference from 5th = $22 - 5 = 17$ days. $17 \div 7$ remainder = 3.
Tuesday + 3 = Friday.

Answer: Friday

Example 10: How many leap years are there in 400 years?

Solution: $400 \div 4 = 100$, but century years 100, 200, 300 are not leap years, so $100 - 3 = 97$.

Answer: 97

Example 11: If today is Wednesday, what day will it be after 65 days?

Solution: $65 \div 7$ remainder = 2. Wednesday + 2 = Friday.

Answer: Friday

Example 12: If two days after tomorrow is Friday, what day was yesterday?

Solution: 3 days from today is Friday, meaning today is Tuesday. Yesterday was Monday.

Answer: Monday

Example 13: What is the maximum number of Mondays in a year?

Solution: An ordinary year has 52, and a leap year can have a maximum of 53 Mondays.

Answer: 53

Example 14: How many odd days are there in 100 years?

Solution: 76 ordinary years and 24 leap years give $76 \times 1 + 24 \times 2 = 124$ or $124 \div 7$ remainder 5.

Answer: 5 odd days

Example 15: If the 3rd day of a month is Tuesday, what day will be 5 days before the 21st of that month?

Solution: 5 days before 21st = 16th. Difference from 3rd = $16 - 3 = 13$ days. $13 \div 7$ remainder 6.

Tuesday + 6 = Monday.

Answer: Monday

Part 3: Practice Questions (25 Questions)

Q1. How many days are there in an ordinary year?

- (A) 365 days (B) 366 days (C) 364 days (D) 367 days

Q2. Which of the following is a leap year?

- (A) 1990 (B) 2012 (C) 1982 (D) 2018

Q3. To check for a century leap year, the year is divided by:

- (A) 4 (B) 100 (C) 400 (D) 8

Q4. How many odd days are there in a leap year?

- (A) 0 (B) 1 (C) 2 (D) 3

Q5. If 1st January 2008 was Tuesday, what day will it be on 1st January 2009?

- (A) Wednesday (B) Thursday (C) Friday (D) Monday

Q6. How many odd days are there in 100 years?

- (A) 3 (B) 5 (C) 1 (D) 0

Q7. If today is Monday, what day will it be after 61 days?

- (A) Wednesday (B) Thursday (C) Saturday (D) Sunday

Q8. The calendar for the year 2015 will be the same for which year?

- (A) 2020 (B) 2021 (C) 2026 (D) 2022

Q9. If the 14th day of a month is Friday, what day will be the 28th of that month?

- (A) Friday (B) Saturday (C) Thursday (D) Sunday

Q10. How many leap years are there in 400 years?

- (A) 100 (B) 97 (C) 96 (D) 99

Q11. If day before yesterday was Sunday, when will Thursday be?

- (A) Today (B) Tomorrow (C) Day after tomorrow (D) After 3 days

Q12. Which of the following can be the last day of a century?

- (A) Tuesday (B) Friday (C) Saturday (D) Wednesday

Q13. Which of the following century years is a leap year?

- (A) 1700 (B) 1800 (C) 2000 (D) 2100

Q14. What is the maximum number of days in the month of February?

- (A) 28 (B) 29 (C) 30 (D) 31

Q15. If the first day of an ordinary year is Monday, what will be the last day?

- (A) Monday (B) Tuesday (C) Sunday (D) Wednesday

Q16. How many odd days are there in 200 years?

- (A) 5 (B) 3 (C) 1 (D) 0

Q17. If the day after tomorrow is Wednesday, what was yesterday?

- (A) Sunday (B) Saturday (C) Friday (D) Monday

Q18. How many odd days are there in 300 years?

- (A) 1 (B) 3 (C) 5 (D) 0

Q19. The calendar for the year 2020 will be the same for which year?

- (A) 2048 (B) 2026 (C) 2031 (D) 2025

Q20. If 5th of a month is Monday, what day will be 3 days after the 20th of that month?

- (A) Friday (B) Saturday (C) Tuesday (D) Wednesday

Q21. How many weeks are there in a year?

- (A) 50 (B) 51 (C) 52 (D) 53

Q22. If the first day of a leap year is Wednesday, what will be the last day?

- (A) Wednesday (B) Thursday (C) Friday (D) Saturday

Q23. How many leap years are there in 50 years?

- (A) 12 (B) 13 (C) 11 (D) 10

Q24. If today is Saturday, what day will it be after 100 days?

- (A) Monday (B) Sunday (C) Tuesday (D) Thursday

Q25. How many odd days are there in 400 years?

- (A) 0 (B) 1 (C) 3 (D) 5

Answer Key for Practice Questions

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
Q1	(A) 365 days	Q8	(C) 2026	Q15	(A) Monday	Q22	(B) Thursday
Q2	(B) 2012	Q9	(A) Friday	Q16	(B) 3	Q23	(A) 12
Q3	(C) 400	Q10	(B) 97	Q17	(A) Sunday	Q24	(A) Monday
Q4	(C) 2	Q11	(B) Tomorrow	Q18	(A) 1	Q25	(A) 0
Q5	(C) Friday	Q12	(B) Friday	Q19	(A) 2048	-	-
Q6	(B) 5	Q13	(C) 2000	Q20	(B) Saturday	-	-
Q7	(A) Wednesday	Q14	(B) 29 days	Q21	(C) 52	-	-